

Delays in cancer detection using 2 and 4-year screening intervals for prostate cancer screening with initial prostate specific antigen less than 2 ng/ml.

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The American Urological Association and American Cancer Society advocate annual screening with serum prostate specific antigen (PSA) and digital rectal examination starting at age 50 years in the general population and earlier in men at high risk. Some groups have suggested that screening at 2 or 4-year intervals may be sufficient in men with initial PSA 2 ng/ml or less. We reviewed the records of men enrolled in a PSA and digital rectal examination based prostate cancer screening study to determine the extent to which the diagnosis of cancer would have been delayed using a 2 or 4-year screening interval. We evaluated 18,140 volunteers in a prostate cancer screening study in whom PSA was less than 2 ng/ml at initial screening and who were screened at 6-month to 1-year intervals for up to 8 years. We evaluated the cancers detected in these intervals to determine the possible delay in cancer diagnosis that would occur using prolonged screening intervals. We report the overall cancer detection rate, clinical and pathological tumor stage, and Gleason grade of the cancers detected. Excluding 70 men in whom prostate cancer was detected at initial evaluation 2.0% had prostate cancer detected during the next 8 years (mean 21.6 cancers per 6 months, median 20, range 12 to 33). Using a hypothetical 2-year screening interval cancer detection 62% would have been delayed by 4 to 20 months. Using a hypothetical 4-year screening interval cancer detection would have been delayed in 77% of men by 4 to 44 months. Of the tumors detected 100% were clinically localized, 77% were pathologically organ confined and 29% had a Gleason score of 7 or greater. The 2 or 4-year PSA screening interval in men with initial serum PSA less than 2 ng/ml would result in substantial delays in prostate cancer detection. To our knowledge the extent to which these delays would affect treatment outcomes is undetermined.

American Urological Association (AUA) guideline on prostate cancer detection: process and rationale.

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To review the process and rationale for the American Urological Association (AUA) guideline on prostate cancer detection. The AUA guideline on detection of prostate cancer involved a systematic literature review of >300 studies that evaluated outcomes important to patients (prostate cancer, incidence/mortality, health-related quality of life, diagnostic accuracy and harms of testing). A multidisciplinary panel interpreted the evidence and formulated statements to assist the urologist and the asymptomatic average-risk man in decision-making about prostate cancer detection. Other than prostate-specific antigen (PSA)-based prostate cancer screening, there was no evidence to address the outcomes of interest to patients. The strongest evidence that benefits may outweigh harms was in men aged 55-69 years undergoing PSA-based screening. This led the panel to recommend shared decision-making for these men at average risk, but recommend against routine screening for other age groups at average risk. Further, to reduce the harms associated with screening (false positive tests, over diagnosis, over treatment), the panel recommended against annual screening for those who choose to be screened. A panel under the auspices of the AUA recommended shared decision-making for the average risk asymptomatic man aged 55-69 years considering PSA-based screening for prostate cancer detection.

Trends in prostate cancer incidence and mortality to monitor control policies in a northeastern Brazilian state.

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Prostate cancer differently affects different regions of the world, displaying higher rates in more developed areas. After the implementation of prostate-specific antigen (PSA) testing, several studies described rising rates globally, but it is possible that indolent lesions are being detected given the lack of changes in mortality data. The Brazilian government recommends against PSA screening in the male population regardless of age, but the Urology Society issued a report recommending that screening should start at 50 years old for certain men and for those aged ≥ 75 years with a life expectancy exceeding 10 years. In this study, we examined the incidence and mortality rates of invasive prostate cancer over time in the Sergipe state of Brazil. The databases of the Aracaju Cancer Registry and Mortality Information System were used to calculate age-standardized rates for all prostate tumors (International Classification of Diseases 10th edition: C61 and D07.5) in the following age ranges: 20-44, 45-54, and ≥ 65 years. We identified 3595 cases of cancer, 30 glandular intraepithelial high-grade lesions, and 3269 deaths. Using the Joinpoint Regression Program, we found that the incidence of prostate cancer dramatically increased over time until the mid-2000s for all age groups, after which the rates declined. Prostate cancer mortality rates increased until 2005, followed by a non-significant annual percent change of 22.0 in 2001-2005 and a stable rate thereafter. We noticed that the increases and decreases of the incidence rates of prostate cancer were associated with the screening recommendations. Meanwhile, the increased mortality rates did not appear to be associated with decreased PSA testing; instead, they were linked to the effects of age and improvements in identification of the cause of death. Thus, we do not believe a PSA screening program would benefit the population of this study.

The epidermal growth factor receptor variant type III mutation frequently found in gliomas induces astrogenesis in human cerebral organoids.

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The epidermal growth factor receptor variant type III (EGFRvIII) is the most common mutation of EGFR in glioblastoma multiforme (GBM) and is found in approximately 25% of all GBMs. Intriguingly, EGFRvIII is mostly found in GFAP+ astrocytic tumour cells in the brain, suggesting connection of EGFRvIII to astrogenesis. In this study, we explored whether EGFRvIII mutation facilitates astrogenesis in human development setting. Using CRISPR-Cas9, we generated EGFRvIII mutations in H9-hESCs. Wild type (wt) H9-hESCs were used as an isogenic control. Next, we generated cerebral organoids using the wt and EGFRvIII-hESCs and examined the astrogenic differentiation of the brain organoids. EGFRvIII-organoids showed abundant astrocytes (GFAP+ , S100 β +), while no astrocytes were detected in wt hESC-derived organoids at day 49. On the contrary, TUJ1+ neurons were more abundant in the wt-organoids than the EGFRvIII-organoids. This result suggested that constitutively active EGFRvIII promoted astrogenesis at the expense of neurogenesis. In addition, the EGFRvIII-organoids were larger in size and retained more Ki67+ cells than wt-organoids, indicating enhanced cell proliferation by the mutation. The EGFRvIII-organoids displayed massive apoptotic cell death after treatment with temozolomide and hence, could be used for evaluation of anti-GBM drugs. EGFRvIII mutation-induced astrogenesis and massive cell proliferation in a human brain development model. These results provide us new insights into the mechanisms relating EGFRvIII mutation-mediated gliogenesis and gliomagenesis.

Changes in the EGFR amplification and EGFRvIII expression between paired primary and recurrent glioblastomas.

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The efficacy of novel targeted therapies is often tested at the time of tumor recurrence. However, for glioblastoma (GBM) patients, surgical resections at recurrence are performed only in a minority of patients; therefore, molecular data are predominantly derived from the initial tumor. Molecular data of the initial tumor for patient selection into personalized medicine trials can therefore be used only when the specific genetic change is retained in the recurrent tumor. In this study we determined whether EGFR amplification and expression of the most common mutation in GBMs (EGFRvIII) is retained at tumor recurrence. Because retention of genetic changes may be dependent on the initial treatment, we only used a cohort of GBM samples that were uniformly treated according to the current standard of care (ie, chemo-irradiation with temozolomide). Our data show that, in spite of some quantitative differences, the EGFR amplification status remains stable in the majority (84%) of tumors evaluated. EGFRvIII expression remained similar in 79% of GBMs. However, within the tumors expressing EGFRvIII at initial diagnosis, approximately one-half lose their EGFRvIII expression at tumor recurrence. The relative stability of EGFR amplification indicates that molecular data obtained in the primary tumor can be used to predict the EGFR status of the recurrent tumor, but care should be taken in extrapolating EGFRvIII expression from the primary tumor, particularly when expressed at first diagnosis.

Prognostic value of epidermal growth factor receptor in patients with glioblastoma multiforme.

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Glioblastoma multiforme (GBM) frequently involves amplification and alteration of the epidermal growth factor receptor (EGFR) gene, resulting in overexpression of varied mutations, including the most common mutation, EGFRvIII, as well as wild-type EGFR (EGFRwt). To test the prognostic value of EGFR, we retrospectively analyzed the relationship between treatment outcomes and the EGFR gene in 87 newly diagnosed adult patients with supratentorial GBM enrolled in clinical trials. The EGFR gene status was assessed by Southern blots and EGFR expression by immunohistochemistry using three monoclonal antibodies (EGFR.25 for EGFR, EGFR.113 for EGFRwt, and DH8.3 for EGFRvIII). EGFR amplification was detected in 40 (46%) of the 87 GBM patients; in 39 (97.5%) of these, EGFR was overexpressed. On the other hand, in 46 of 47 patients without EGFR amplification (97.9%), no EGFR overexpression was present. There was a close correlation between EGFR amplification and EGFR overexpression ($P < 0.0001$). EGFRwt was overexpressed in 27 of the 40 (67.5%) patients with, and in none without, EGFR amplification ($P < 0.0001$). Similarly, EGFRvIII was overexpressed in 18 (45.0%) of 40 patients with and in 4 (8.5%) of 47 patients without EGFR amplification ($P < 0.0001$). The finding that 8 (20%) of the patients with EGFR amplification/EGFR overexpression manifested overexpression of neither EGFRwt nor EGFRvIII indicates that they overexpressed other types of EGFR. Multivariate analysis demonstrated that EGFR amplification was an independent, significant, unfavorable predictor for overall survival (OS) in all patients ($P = 0.038$, HR = 1.67). With respect to the relationship of age to EGFR prognostication, the EGFR gene status was a more significant prognosticator in younger patients, particularly in those <60 years ($P = 0.0003$, HR = 3.15), whereas not so in older patients. EGFRvIII overexpression, on the other hand, was not predictive for OS. However, in patients with EGFR amplification, multivariate analysis revealed that EGFRvIII overexpression was an independent, significant, poor prognostic factor for OS ($P =$

0.0044, HR = 2.71). This finding indicates that EGFRvIII overexpression in the presence of EGFR amplification is the strongest indicator of a poor survival prognosis. In GBM patients, EGFR is of significant prognostic value for predicting survival, and the overexpression of EGFRvIII with amplification plays an important role in enhanced tumorigenicity.

Maintenance of EGFR and EGFRvIII expressions in an in vivo and in vitro model of human glioblastoma multiforme.

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Glioblastoma multiforme (GBM) is the most common, and most aggressive primary brain tumor among adults. A vast majority of the tumors express high levels of the epidermal growth factor receptor (EGFR) as a consequence of gene amplification. Furthermore, gene amplification is often associated with mutation of EGFR, and the constitutive activated deletion variant EGFRvIII is the most common EGFR mutation found in GBM. Activated EGFR signaling, through overexpression and/or mutation, is involved in increased tumorigenic potential. As such, EGFR is an attractive target for GBM therapy. However, clinical studies with EGFR inhibitors have shown inconsistent results, and as such, further knowledge regarding the role of EGFR and EGFRvIII in GBM is needed. For this, an appropriate in vivo/in vitro tumor model is required. Here, we report the establishment of an experimental GBM model in which the expressions of EGFR and EGFRvIII are maintained both in xenograft tumors growing subcutaneously on mice and in cell cultures established in stem cell conditions. With this model it will be possible to further study the role of EGFR and EGFRvIII, and response to targeted therapy, in GBM.

The EGFR variant III mutant as a target for immunotherapy of glioblastoma multiforme.

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In epithelial tumors, the epidermal growth factor receptor (EGFR) controls key signaling pathways responsible for growth, proliferation, migration, and survival of tumor cells. The epidermal growth factor receptor variant III (EGFRvIII) is the most common EGFR mutation that occurs in up to 30% of high-grade gliomas especially glioblastoma multiforme (GBM). EGFRvIII arises from the deletion of exon 2-7 that leads to the formation of the constitutively activated mutant receptor incapable of binding any known ligand. EGFRvIII-expressing cells are resistant to EGFR inhibitors and therefore take advantage in survival. Furthermore, EGFRvIII expression in tumors is often correlates with poor prognosis. Indeed, EGFRvIII targeting is of great therapeutic value in order to inhibit GBM progression and invasion. The presence of the unique glycine site in EGFRvIII provides an option to develop EGFRvIII-specific monoclonal antibodies (MAbs). These antibodies are used for detection of EGFRvIII and immunotherapy. Peptide CDX-110 derived from EGFRvIII has high immunogenicity. This peptide was used for the development of anti-EGFRvIII MAbs with improved specificity to the mutant receptor, EGFRvIII-specific dendritic cell vaccine, and CDX-110-KLH peptide vaccine. These immune reagents were successfully tested in preclinical and clinical studies for GBM treatment.

The EGFRvIII variant in glioblastoma multiforme.

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Glioblastoma multiforme (GBM) is the most common brain tumour and has the worst prognosis. Epidermal growth factor receptor (EGFR) gene amplification, mutation and re-arrangement (all of which enhance tumour growth, survival, progression and resistance to therapy) are frequently observed in primary GBM. The most common EGFR variant in GBM, the EGFRvIII, is characterised by a deletion of 267 amino acids in the extracellular domain, leading to a receptor which is unable to bind ligand yet is constitutively active. Together with its impaired internalisation and degradation, the EGFRvIII enhances the tumourigenic potential of GBM by activating and sustaining mitogenic, anti-apoptotic and pro-invasive signalling pathways. This EGFRvIII-mediated enhanced tumourigenicity combined with the lack of EGFRvIII expression in normal tissue makes it an ideal candidate for targeted therapy. This review summarizes the current knowledge about the role of EGFRvIII in GBM and discusses therapeutic agents targeting EGFRvIII that are being evaluated as treatments for GBM.

Predominant expression of mutant EGFR (EGFRvIII) is rare in primary glioblastomas.

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EGFR amplification is a frequent genetic alteration in primary (de novo) glioblastomas, and is often associated with structural alterations. Most common is variant III (EGFRvIII), which results from a non-random 801 bp in-frame deletion of exons 2 to 7 of the EGFR gene. We assessed amplification and overexpression of EGFRvIII and wild-type EGFR in 30 glioblastoma biopsies. Immunohistochemically, EGFR overexpression was observed in 20 (67%) of 30 glioblastomas. Eight (27%) cases also showed immunoreactivity to an EGFRvIII antibody. In 6 of these cases, the pattern of EGFR and EGFRvIII overexpression was compared in serial sections: In 4 cases, areas with immunoreactivity to EGFRvIII largely coincided with wild-type EGFR expression. In the other 2 cases, the areas immunoreactive to EGFRvIII were significantly less extensive than EGFR-positive areas. To assess whether EGFRvIII is predominantly amplified in tumors with concurrent wild-type EGFR amplification, we carried out real-time quantitative PCR using 2 sets of primers located in exon 2 and intron 15 of the EGFR gene. A > 5-fold ratio of relative copy numbers between intron 15 (present both in wild-type EGFR and EGFRvIII) and exon 2 (present only in wild-type EGFR, but missing in EGFRvIII) suggested predominant amplification of EGFRvIII in only 3 (10%) of 30 glioblastomas. The observation that intratumoral wild-type EGFR overexpression is often more extensive and that predominant amplification of EGFRvIII is a rare event would limit the effectiveness of therapeutic approaches based on selective targeting of EGFRvIII.